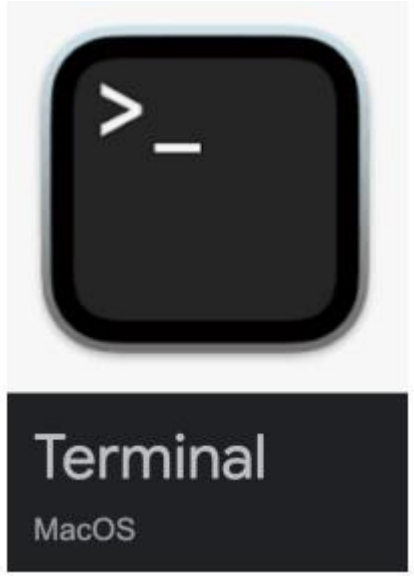
LAB1

For Mac user

1. Open Terminal



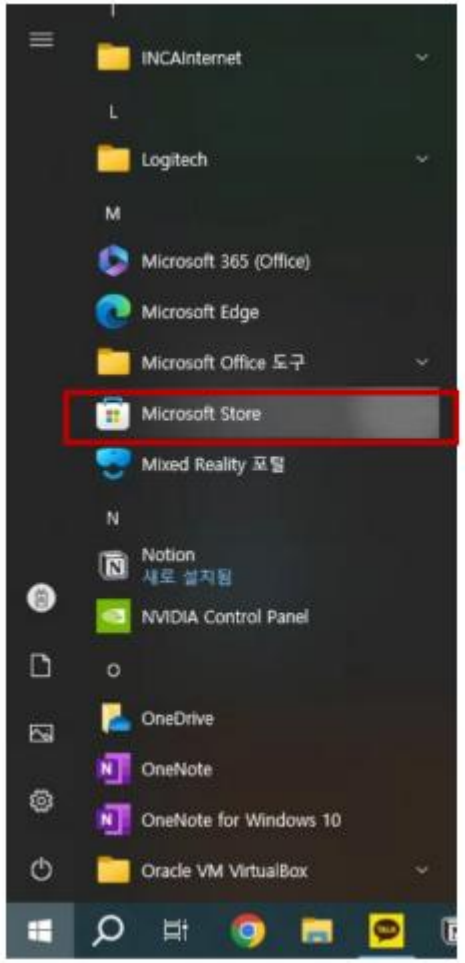
2. Install Homebrew
(https://brew.sh/)

3. Install **gcc-13.3**
>brew install gcc

4. Check gcc version
> gcc --version

For Window user

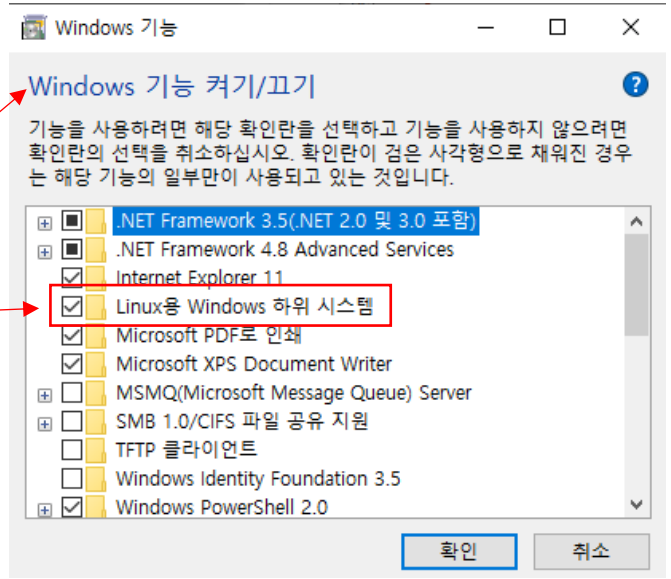
1. Open Microsoft store



2. Download Ubuntu 24.04.1 LTS

For Window user

3. Search “Turn windows features on or off(Windows 기능 켜기/끄기)”
4. Turn on the option “Window subsystem for Linux”
5. Re-start your computer
6. Execute Ubuntu as administrator
7. Install **gcc-13.3**
   > sudo apt update
   > sudo apt install build-essential
8. Access ubuntu folder
   > explorer.exe .



Linux Practice

1. Create a directory

```
$ mkdir [directory name]
```

2. Move to project directory

```
$ cd [directory name]
```

```
mykim@DESKTOP-PVGHFT5:~/DS2026$ mkdir practice
mykim@DESKTOP-PVGHFT5:~/DS2026$ ls
practice
mykim@DESKTOP-PVGHFT5:~/DS2026$ cd practice
mykim@DESKTOP-PVGHFT5:~/DS2026/practice$
```

Linux Practice

3-1. Create a c file

```
$ vi [file name].c
```

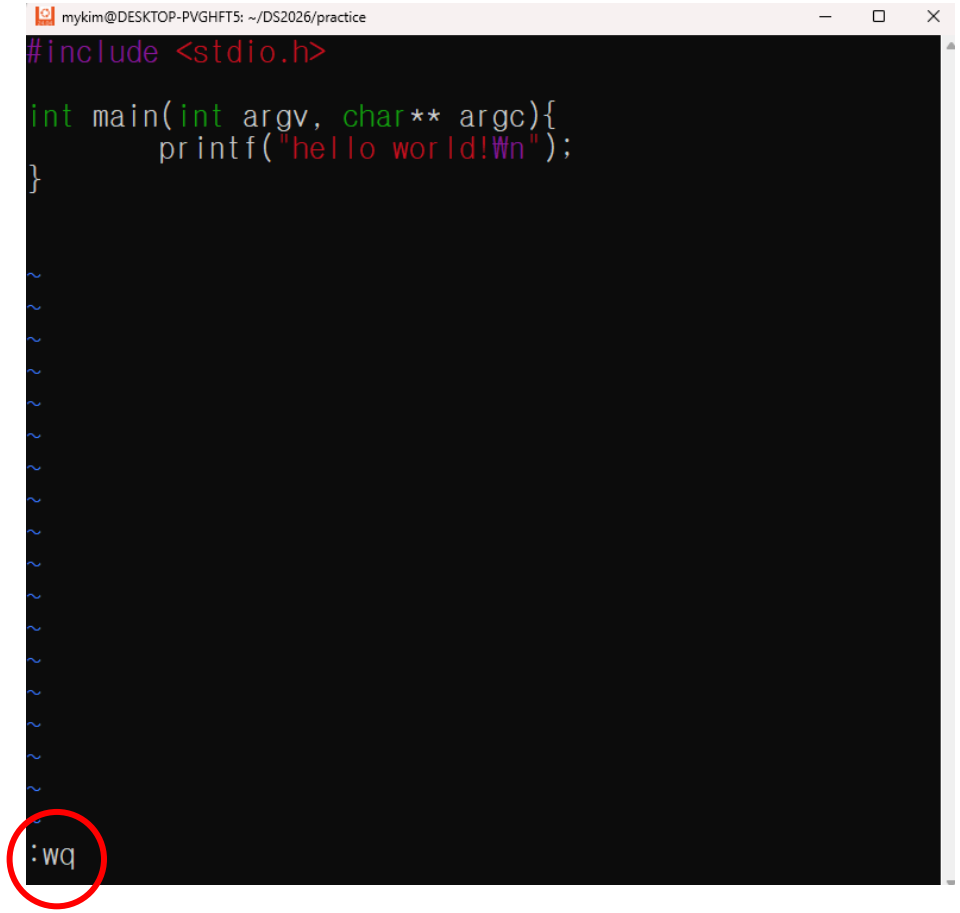
3-2. Type **'i'** to switch to insert mode.

3-3. Type your code

Linux Practice

3-4. Save the file

press [ESC] ▶ type ":wq"



Linux Practice

4. Compile and execute

$ gcc –o [object file name] [code file name]

$ ./[object file name]

```
mykim@DESKTOP-PVGHFT5:~/DS2026/practice$ ls
hello_world.c
mykim@DESKTOP-PVGHFT5:~/DS2026/practice$ gcc -o hello_world hello_world.c
mykim@DESKTOP-PVGHFT5:~/DS2026/practice$ ls
hello_world  hello_world.c
mykim@DESKTOP-PVGHFT5:~/DS2026/practice$ ./hello_world
hello world!
mykim@DESKTOP-PVGHFT5:~/DS2026/practice$
```

Linux Practice

5. Move the file

$ mv [old location] [new location]

```
mykim@DESKTOP-PVGHFT5:~/DS2026/practice$ ls
hello_world  hello_world.c
mykim@DESKTOP-PVGHFT5:~/DS2026/practice$ mv ./hello_world.c ../
mykim@DESKTOP-PVGHFT5:~/DS2026/practice$ ls
hello_world
mykim@DESKTOP-PVGHFT5:~/DS2026/practice$ cd ..
mykim@DESKTOP-PVGHFT5:~/DS2026$ ls
hello_world.c  practice
mykim@DESKTOP-PVGHFT5:~/DS2026$
```

../ : Parent directory

./ : Current directory

Submission Rules

- You should finish the practice (Lab 1) during the lab session and submit it to the **LMS**. You can leave the room after confirming with the TA.
- File name : {student_id}_p1.c (2024XXXXXX_p1.c)
- **Submit only your source file (.c). Do not submit object files or executable files!**
- **The output format must be identical to the given example. Any deviation from the specified format may result in point deductions.**

Lab1. Gugudan printing program with dynamic memory allocation

1. Receive the multiplication table number from the user
2. Store the number using **dynamically allocated memory**
3. Print the multiplication table for the given number
4. Release the allocated memory after use.

[Error handling]

If the user inputs a non-numeric value, the program must print the "Invalid input. Please enter a number!" and terminate the program.

Lab1. Gugudan printing program with dynamic memory allocation

EX 1)

```
mykim@DESKTOP-PVGHFT5:~/DS2026/lab1$ gcc -o 2025123456_p1 2025123456_p1.c
mykim@DESKTOP-PVGHFT5:~/DS2026/lab1$ ./2025123456_p1
Enter the multiplication table number: 5
== 5 times table ==
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
```

Test your program with several numbers to confirm that the multiplication table prints correctly for all cases!!

EX 2)

```
mykim@DESKTOP-PVGHFT5:~/DS2026/lab1$ ./2025123456_p1
Enter the multiplication table number: a
Invalid input. Please enter a number!
mykim@DESKTOP-PVGHFT5:~/DS2026/lab1$
```